Eric Boittier, Ph.D.

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Arlesheim, BL 4144, Switzerland

Post Doctoral Scientist in Artificial Intelligence building Machine Learning Models and Scientific Software Solutions for Research in the Natural Sciences.

Education

Doctor of Philosophy, Chemistry — Basel, CH
University of Basel — June 2025

Thesis: *Molecular Deep Learning for Quantitative Simulations and Electrostatic Models.*

Bachelor of Advanced Science (Honours I), Chemistry — Brisbane, AU
The University of Queensland — January 2020

Thesis: *Development of Computational Tools for the Rational Design of Glycosaminoglycan Mimetics.*

Research Experience

Researcher — Basel, CH
University of Basel — Department of Chemistry — January 2020 - Present

- Deep learning for graph-structured scientific data and computational chemistry.
- Design, training, and validation of machine-learned potentials and distributed charge models.
- Top performance in the HyDRA computational vibrational spectroscopy challenge (Physical Chemistry Chemical Physics, 2023).

Research Assistant — Brisbane, AU
Translational Research Institute, Cancer and Aging Research Program — January 2019 - December 2020

- Interdisciplinary collaboration on small-molecule and biomolecular modelling (Schrödinger suite, OpenMM).
- Structure- and ligand-based drug design; data pipelines in Python and R.

Invited presentations

1. **Eric Boittier**. Bridging machine learning force fields with anisotropic electrostatic models. *Swiss Chemical Society, Machine Learning Group*, Lausanne, CH; Talk.
2. **Eric Boittier**. Mixing machine-learned and empirical energy functions. *apoCHARMM Meeting, NIH*, Boston, US (online); Talk.

Service

Journal of Chemical Physics (AIP Publishing), **Peer reviewer** — January 2020 - Present
Manuscript review

Skills

Programming: Python | Julia | R | Fortran | C++ | OpenMPI | CUDA
Frontend Development: Typescript | JavaScript | React | Tailwind CSS
Machine Learning: JAX | PyTorch | TensorFlow | scikit-learn | Weights & Biases
Engineering: Git | CI/CD | Docker | Slurm
Data Science: SnakeMake | SQL | Polars | Pandas
Large Language Models: Unsloth | Hugging Face | LiteLLM | LangChain | OpenClaw

Publications

1. JingChun Wang, Meenu Upadhyay, **Eric D Boittier**, Kham Lek Chaton, Valerii Andreichev, Mike Devereux, Raidel Martin Barrios, Shimoni Patel, Sena Aydin, Kai Töpfer. Cluster Models for Next-Generation, Machine-Learning-Based Energy Functions for Molecular Simulations. *Small Structures* **2026**, 7 (2), e202500656.
2. **Eric D Boittier**, Markus Meuwly. Efficient, Equivariant Predictions of Distributed Charge Models. . Preprint available on *arXiv*, January 01, 2026. DOI: 10.48550/arXiv.2602.07234
3. Kham Lek Chaton, **Eric D Boittier**, Mike Devereux, Markus Meuwly. Explicit, Machine-Learned Two-Body Potentials for Molecular Simulations. . Preprint available on *arXiv*, January 01, 2026. DOI: 10.48550/arXiv.2603.14466
4. **Eric D Boittier**, Silvan Kaser, Markus Meuwly. Roadmap to CCSD (T)-Quality Machine-Learned Potentials for Condensed Phase Simulations. *Journal of Chemical Theory and Computation* **2025**, 21 (18), 8683-8698.
5. **Eric D Boittier**, Silvan Käser, Markus Meuwly. Towards Large-Scale Condensed Phase Simulations using Machine Learned Energy Functions. . Preprint available on *arXiv*, January 01, 2025. DOI: 10.48550/arXiv.2506.23272
6. Wonmuk Hwang, Steven L Austin, Arnaud Blondel, **Eric D Boittier**, Stefan Boresch, Matthias Buck, Joshua Buckner, Amedeo Caflisch, Hao-Ting Chang, Xi Cheng. CHARMM at 45: Enhancements in accessibility, functionality, and speed. *The Journal of Physical Chemistry B* **2024**, 128 (41), 9976-10042.
7. Kai Topfer, **Eric Boittier**, Mike Devereux, Andrea Pasti, Peter Hamm, Markus Meuwly. Force fields for deep eutectic mixtures: application to structure, thermodynamics and 2D-Infrared spectroscopy. *The Journal of Physical Chemistry B* **2024**, 128 (44), 10937-10949.
8. **Eric Boittier**, Kai Töpfer, Mike Devereux, Markus Meuwly. Kernel-based minimal distributed charges: A conformationally dependent ESP-model for molecular simulations. *Journal of Chemical Theory and Computation* **2024**, 20 (18), 8088-8099.
9. Mike Devereux, **Eric D Boittier**, Markus Meuwly. Systematic improvement of empirical energy functions in the era of machine learning. *Journal of Computational Chemistry* **2024**, 45 (22), 1899-1913.
10. Haydar Taylan Turan, **Eric Boittier**, Markus Meuwly. Interaction at a distance: Xenon migration in Mb. *The Journal of Chemical Physics* **2023**, 158 (12).
11. Tabassum Khair Barbhuiya, Mark Fisher, **Eric D Boittier**, Emma Bolderson, Kenneth J O'Byrne, Derek J Richard, Mark Nathaniel Adams, Neha S Gandhi. Structural investigation of CDCA3-Cdh1 p rotein–protein interactions using in vitro studies and molecular dynamics simulation. *Protein Science* **2023**, 32 (3), e4572.
12. Taija L Fischer, Margarethe Bödecker, Sophie M Schweer, Jennifer Dupont, Valéria Lepère, Anne Zehnacker-Rentien, Martin A Suhm, Benjamin Schröder, Tobias Henkes, Diego M Andrada. The first HyDRA challenge for computational vibrational spectroscopy. *Physical Chemistry Chemical Physics* **2023**, 25 (33), 22089-22102.
13. **Eric D Boittier**, Mike Devereux, Markus Meuwly. Molecular dynamics with conformationally dependent, distributed charges. *Journal of Chemical Theory and Computation* **2022**, 18 (12), 7544-7554.
14. **Eric D Boittier**, Norbert Wimmer, Alexandria K Harris, Mark A Schembri, Vito Ferro. Synthesis of a Gal-ß-(1? 4)-Gal disaccharide as a ligand for the fimbrial adhesin UcaD. *Australian Journal of Chemistry* **2022**, 76 (1), 30-36.
15. Luis Itza Vazquez-Salazar, **Eric D Boittier**, Markus Meuwly. Uncertainty quantification for predictions of atomistic neural networks. *Chemical science* **2022**, 13 (44), 13068-13084.
16. Katrina Kildey, Neha S Gandhi, Katherine B Sahin, Esha T Shah, **Eric Boittier**, Pascal HG Duijf, Christopher Molloy, Joshua T Burgess, Sam Beard, Emma Bolderson. Elevating CDCA3 levels in non-small cell lung cancer enhances sensitivity to platinum-based chemotherapy. *Communications Biology* **2021**, 4 (1), 638.
17. Sarah-Louise Ryan, Keyur A Dave, Sam Beard, Martina Gyimesi, Matthew McTaggart, Katherine B Sahin, Christopher Molloy, Neha S Gandhi, **Eric Boittier**, Connor G O'Leary. Identification of proteins deregulated by platinum-based chemotherapy as novel biomarkers and therapeutic targets in non-small cell lung cancer. *Frontiers in oncology* **2021**, 11, 615967.

18. Luis Itza Vazquez-Salazar, **Eric D Boittier**, Oliver T Unke, Markus Meuwly. Impact of the characteristics of quantum chemical databases on machine learning prediction of tautomerization energies. *Journal of chemical theory and computation* **2021**, 17 (8), 4769-4785.

19. Silvan Kaser, **Eric D Boittier**, Meenu Upadhyay, Markus Meuwly. Transfer learning to CCSD (T): Accurate anharmonic frequencies from machine learning models. *Journal of Chemical Theory and Computation* **2021**, 17 (6), 3687-3699.

20. **Eric D Boittier**, Yat Yin Tang, McKenna E Buckley, Zachariah P Schuurs, Derek J Richard, Neha S Gandhi. Assessing molecular docking tools to guide targeted drug discovery of CD38 inhibitors. *International journal of molecular sciences* **2020**, 21 (15), 5183.

21. **Eric D Boittier**, Jed M Burns, Neha S Gandhi, Vito Ferro. GlycoTorch Vina: docking designed and tested for glycosaminoglycans. *Journal of Chemical Information and Modeling* **2020**, 60 (12), 6328-6343.

22. **Eric D Boittier**, Neha S Gandhi, Vito Ferro, Deirdre R Coombe. Cross-species analysis of glycosaminoglycan binding proteins reveals some animal models are "More Equal" than others. *Molecules* **2019**, 24 (5), 924.

23. Jed M Burns, **Eric D Boittier**. Pathway Bifurcation in the (4 + 3)/(5 + 2)-Cycloaddition of Butadiene and Oxidopy- rylium Ylides: The Significance of Molecular Orbital Isosymmetry.. *Journal of Organic Chemistry* **2019**, 84 (10), 5997-6005.